

Analysis of Delay in Computer Networks

Your briefing is technically sound and well-aligned with **flow control analysis (Stop-and-Wait, GBN, SR)**. Below is a **clean synthesis + intuition-building expansion**, with diagrams and exam-centric clarifications that make the delay components *operationally obvious*, not just formulaic.

1 What “Network Delay” Really Means

Network delay is the total elapsed time from the moment a sender starts transmitting a packet until the **last bit** of that packet is successfully received at the destination.

It is the **sum of four logically independent delays**:

[

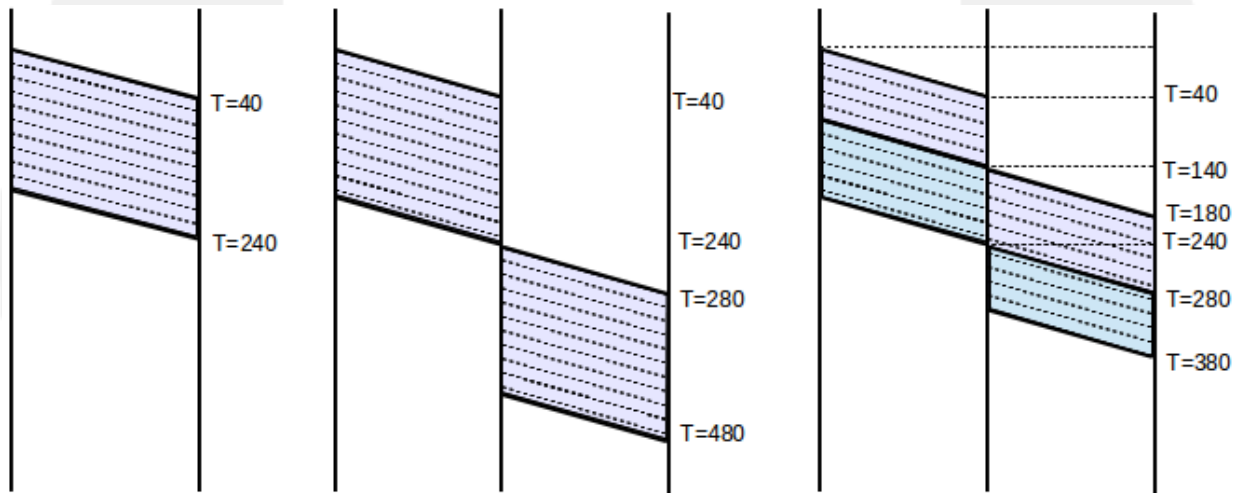
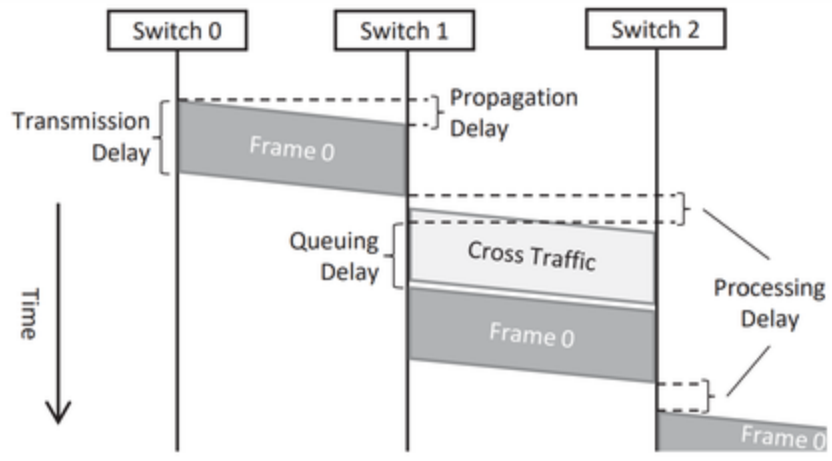
\boxed{

\text{Total Delay}

$T_d + P_d + Q_d + P_{rd}$
}
]

Each term corresponds to a **different physical or logical bottleneck**.

2 The Four Delay Components — With Intuition

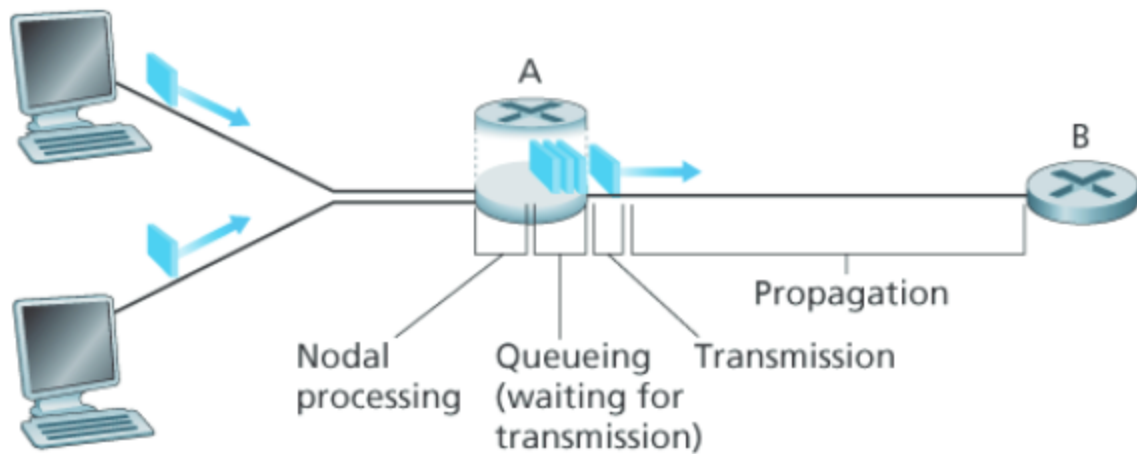


Case 1: one link

Case 3: two links

Case 4: two half-sized packets

Bandwidth delay 200 μsec , per-link propagation delay 40 μsec



2.1 Transmission Delay (T_d)

What it measures

Time required to *push all bits of the packet* onto the link.

Formula

$$T_d = \frac{L}{B}$$

Parameter	Meaning
L	Packet length (bits)
B	Bandwidth (bits/sec)

Key Insight

- This delay exists **even if the receiver is right next to the sender**
- It depends **only on sender speed**, not distance

📌 *Analogy:* Loading passengers onto a bus

📌 *Extreme case:*

Bandwidth = 1 bps, Packet = 100 bits → **100 seconds just to transmit**

2.2 Propagation Delay (P_d)

What it measures

Time taken by a bit to physically travel through the medium.

Formula

$$P_d = \frac{D}{V}$$

Parameter

Meaning

D	Distance (meters)
V	Signal speed ($\approx 2 \times 10^8$ m/s in fiber)

Key Insight

- Independent of packet size
- Independent of bandwidth
- Dominates **long-distance links** (satellite, submarine cables)

📌 *Analogy:* Time for sound to travel from speaker to ear

2.3 Queuing Delay (Q_d)

What it measures

Waiting time in router buffers due to congestion.

Properties

- No closed-form formula
- Depends on:
 - Traffic load
 - Router scheduling
 - Buffer size

Range

$$0 \leq Q_d < \infty$$

📌 *Exam Rule:*

If not explicitly stated → **assume ($Q_d = 0$)**

2.4 Processing Delay (P_{rd})

What it measures

Time taken to inspect and process the packet.

Typical tasks

- Header extraction
- CRC check
- Route lookup
- Buffer assignment

📌 Usually **microseconds or less**

📌 Ignored in most theoretical problems unless specified

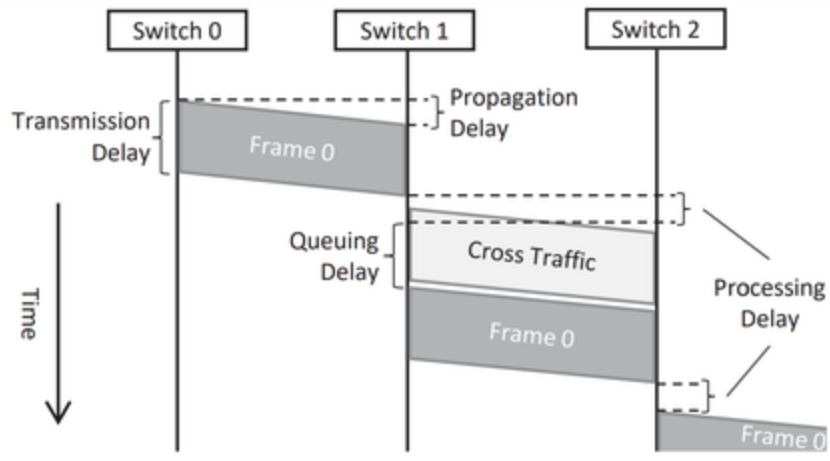
3 Transmission vs Propagation — The Most Tested Concept

Aspect	Transmission Delay	Propagation Delay
Depends on packet size?	✓ Yes	✗ No
Depends on distance?	✗ No	✓ Yes
Depends on bandwidth?	✓ Yes	✗ No
Physical meaning	Loading bits	Moving bits

Transmission = “putting data on the wire”

Propagation = “data moving through the wire”

4 The Pipelining Effect (Why Delays Overlap)



5 Unit Discipline — The Silent Exam Killer

Your emphasis here is **extremely important**.

Data vs Bandwidth Units

Prefix	Data (Base-2)	Bandwidth (Base-10)
K	$2^{10} = 1024$	$10^3 = 1000$
M	2^{20}	10^6
G	2^{30}	10^9

Example (Correct vs Wrong)

Packet = **8 K bits**

Bandwidth = **8 Kbps**

✓ Correct:

$$\left[\frac{8 \times 1024}{8 \times 1000} = 1.024 \sim \text{text{s}} \right]$$

✗ Wrong:

$$\left[\frac{8000}{8000} = 1 \sim \text{text{s}} \right]$$

GATE uses this mistake deliberately.

6 Layer-wise Context of Delay

Layer	Address Type	Delay Contribution
Data Link	MAC (48-bit)	Transmission
Network	IP (32-bit)	Queuing + Routing
Transport	Port (16-bit)	Flow control effects

7 Where Each Delay Dominates (Real-World Insight)

Scenario	Dominant Delay
LAN	Transmission
WAN	Propagation
Satellite (36,000 km)	Propagation
Congested Router	Queuing

Satellite round-trip delay $\approx$ **500 ms**, even at infinite bandwidth.

8 Exam Assumptions (Very Important)

Unless explicitly stated:

- $(Q_d = 0)$
- $(P_{rd} = 0)$

Thus:

```
[
\boxed{
\text{Total Delay} = T_d + P_d
}
]
```

This simplification underpins:

- Stop-and-Wait efficiency
- Sliding Window utilization
- Bandwidth–Delay Product (BDP)

